

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Data Flow Diagrams & User Stories

ShopEZ — MERN Stack E-commerce Application

Phase: Requirement Analysis Phase

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Data Flow Diagrams & User Stories

1. Introduction

This document presents the **Data Flow Diagrams (DFDs)** and formal **User Stories** for the ShopEZ platform. DFDs model the system at two abstraction levels — Context Level (Level 0) and Functional Decomposition (Level 1) — illustrating data flows between actors, processes, and data stores.

2. Level 0 — Context Diagram

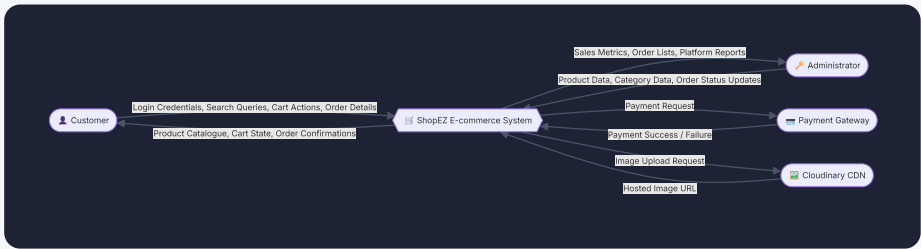


FIGURE 6.1 — LEVEL 0: CONTEXT DIAGRAM (SYSTEM AS BLACK BOX)

3. Level 1 — Authentication Sub-Process

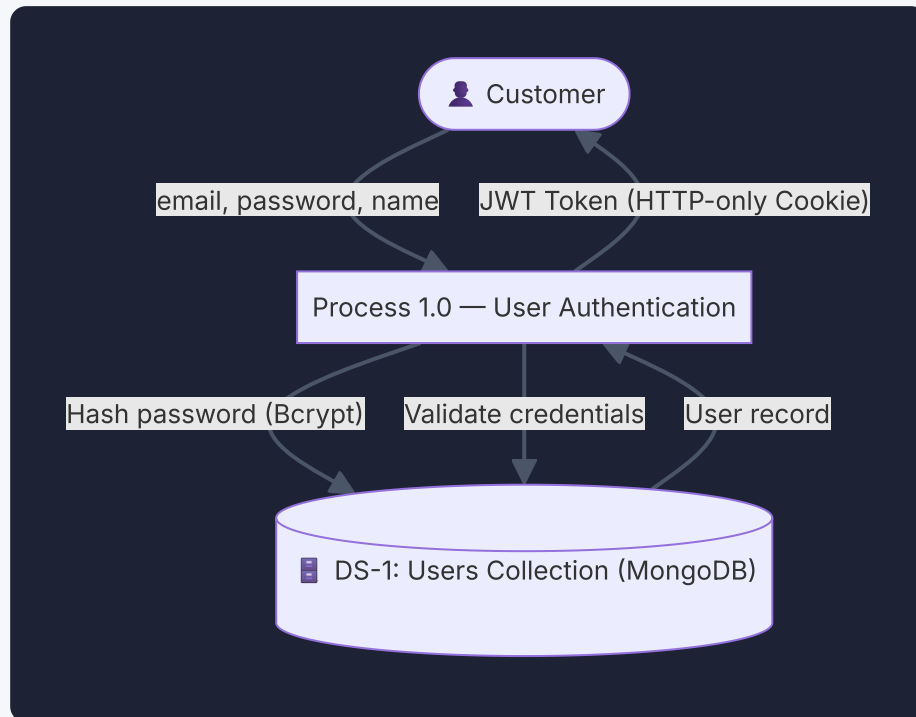


FIGURE 6.2 — LEVEL 1: AUTHENTICATION PROCESS DFD

4. Level 1 — Shopping & Order Placement Sub-Process

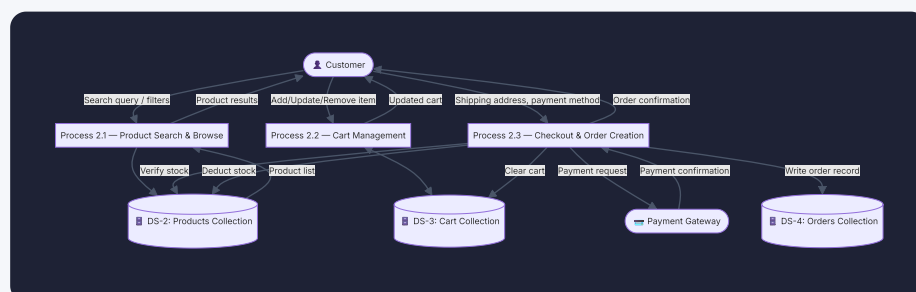


FIGURE 6.3 — LEVEL 1: SHOPPING & ORDER PLACEMENT DFD

5. User Stories

USN	User Type	User Story	Acceptance Criteria	Points	Sprint
USN-1	Customer	As a customer, I can register by entering my name, email, and password.	Account created in DB; redirected to dashboard; duplicate email returns 400.	2	Sprint 1
USN-2	Customer	As a customer, I can log in using my registered email and password.	JWT issued in HTTP-only cookie; incorrect credentials return 401.	2	Sprint 1
USN-3	Customer	As a customer, I can search and filter products by category, price, and rating.	Results accurately reflect filters; update without full page reload.	3	Sprint 1
USN-4	Customer	As a customer, I can add, remove, and update quantities in my cart.	Cart total updates in real time; cart persists across sessions and devices.	3	Sprint 2

USN	User Type	User Story	Acceptance Criteria	Points	Sprint
USN-5	Customer	As a customer, I can complete a guided checkout by entering shipping and payment details.	All three checkout steps navigable; address validated before proceeding.	3	Sprint 2
USN-6	Customer	As a customer, I can securely pay via an integrated payment gateway.	Payment processed; order status changes to "Paid"; stock deducted; confirmation shown.	5	Sprint 2
USN-7	Customer	As a customer who has completed a purchase, I can leave a star rating and review.	Review appears on the product page; aggregate rating recalculated.	2	Sprint 3
USN-8	Admin	As an admin, I can add, edit, and delete product listings with Cloudinary	Changes reflect immediately on storefront; images served via CDN URL.	3	Sprint 1

USN	User Type	User Story	Acceptance Criteria	Points	Sprint
		image uploads.			
USN-9	Admin	As an admin, I can view all orders and update their delivery status.	Status transitions: Processing → Shipped → Delivered; reflected in customer's order history.	2	Sprint 3
USN-10	Admin	As an admin, I can view aggregated sales metrics and monthly revenue charts.	Dashboard shows accurate total revenue, order count, and monthly breakdown.	3	Sprint 3